A One-Way Coupled 2D–3D Framework for Reconstructing the 2011 Tohoku Tsunami

Regional nonlinear shallow-water propagation to Local3D URANS–VOF replay at Kamaishi

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Why this matters

Tsunami-defence assessment needs local impact predictions that are physically linked to an event-scale source. A full 3D simulation from the earthquake source to the coast is not practical for routine design studies, while a purely depth-averaged model cannot resolve local inundation structure, forces, moments or defence interaction.

Poster message. The project builds an auditable one-way hybrid framework: a Regional2D model carries the Tohoku event toward Kamaishi, then Local3D OpenFOAM replay resolves local free-surface impact under the exported forcing.

Aim and objectives

Aim: establish a validated event-reconstruction framework that can later support comparative assessment of coastal-defence configurations.

- Reconstruct coseismic tsunami generation from the 2011 Tohoku event.
- Propagate the wave through a source-to-coast Regional2D corridor.
- Export physically consistent nearshore forcing at a Kamaishi coupling interface.
- Replay that forcing in a Local3D URANS–VOF model for local impact.
- Quantify uncertainty through verification, convergence, calibration and validation gates.

Real-event reconstruction

The Kamaishi delivery case is shown as a physical sequence from regional event context, through the 3D corridor terrain, to the longitudinal bathymetric interpretation at the current 2D–3D extraction interface.

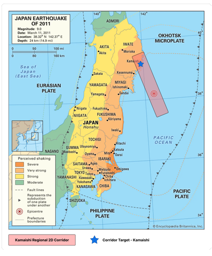

Figure A. Regional 2011 Tohoku event context and Kamaishi Regional2D corridor. Base map adapted from Encyclopaedia Britannica; corridor and Kamaishi annotations added for this study.

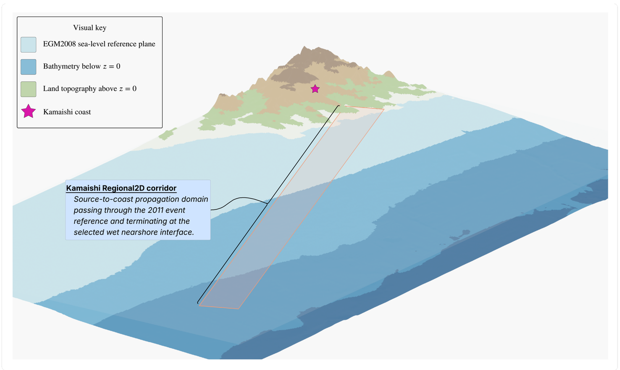

Figure B. Oblique Kamaishi corridor bathymetry/topography. Terrain uses NOAA/NCEI ETOPO 2022, EGM2008, QGIS/GDAL and Blender 5.2 EEVEE with 4× vertical exaggeration.

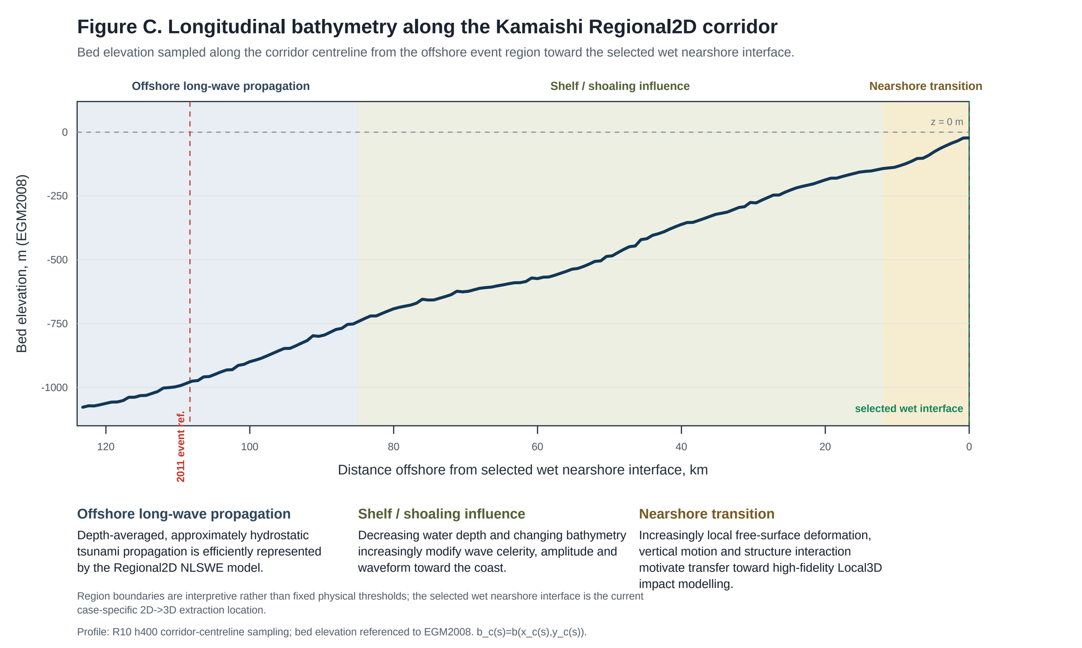

Figure C. Frozen R10 h400 centreline bathymetry with interpretive long-wave, shoaling and nearshore-transition zones; supported by Saito et al. (2014), Fujima et al. (2002) and Takase et al. (2016).

Current status

G6
baseline closed
The theoretical hybrid model and real Kamaishi replay acceptance are complete.

C1A
in progress
Regional convergence evidence exists, but forcing-waveform convergence is not yet qualified.

Next
validation
Calibration and observational validation remain future stages.

Hybrid framework overview

Finite-fault source
coseismic seabed displacement
↓
Terrain and corridor preprocessing
ETOPO-derived Kamaishi delivery geometry
↓
Regional2D nonlinear shallow-water propagation
finite-volume free-surface and momentum fields
↓
Coupling export
nearshore interface histories: $\eta(t), h(t), \bar{\mathbf{u}}(t)$
↓
OpenFOAM Local3D replay
URANS–VOF inlet reconstruction
↓
Impact visualisation, probes, forces, moments
validation and defence assessment

Mathematical model

The Regional2D component solves a depth-averaged nonlinear shallow-water system in conservative form:

$$\partial_t h + \nabla\cdot(h\mathbf{u}) = 0,$$

$$\partial_t(h\mathbf{u}) + \nabla\cdot\left(h\mathbf{u}\otimes\mathbf{u} + \frac{1}{2}gh^2\mathbf{I}\right) = \mathbf{S}_b + \mathbf{S}_f + \mathbf{S}_{eq}.$$

The Local3D replay uses an incompressible two-phase URANS–VOF formulation:

$$\nabla\cdot\mathbf{U} = 0, \qquad \partial_t(\rho\mathbf{U}) + \nabla\cdot(\rho\mathbf{U}\otimes\mathbf{U}) = -\nabla p + \nabla\cdot\tau + \rho\mathbf{g},$$

$$\partial_t\alpha + \nabla\cdot(\alpha\mathbf{U}) + \nabla\cdot(\alpha(1-\alpha)\mathbf{U}_c) = 0.$$

One-way coupling maps Regional2D free-surface elevation and depth-averaged momentum histories to the 3D inlet boundary. Feedback from Local3D to the Regional2D field is outside the current baseline.

Implementation pipeline

Stage	Current implementation role
Source	USGS finite-fault artefact produces coseismic bed displacement.
Terrain	ETOPO-derived conditioning and projected corridor construction.
Regional2D	Repository-native finite-volume propagation and diagnostics.
Coupling	Nearshore interface export of histories and metadata.
Local3D	OpenFOAM Foundation 11 replay generation and execution.
Evidence	JSON, CSV, VTK and figure provenance for reproducibility.

Figure placeholder: OpenFOAM replay visual

Insert final single-panel rendered free-surface or velocity field from the Local3D replay.

single-panel figure slot

Verification and numerical findings

The poster baseline reports the project honestly: verification evidence has advanced, but full convergence, calibration and validation are not complete.

- G6 established the theoretical hybrid model baseline and accepted real Kamaishi Local3D replay evidence.
- C1A Regional2D convergence work corrected a previous fine-mesh construction ambiguity.
- The latest Regional ladder reuses h1000 and adds h900 and h800 full 600 s runs.
- The medium-to-fine forcing waveform comparison still exceeds the 2% qualification target.

Current conclusion: Regional forcing-waveform convergence is **not yet qualified**; no production mesh or timestep policy is selected from C1A.

Key output slots

Figure placeholder: Regional forcing waveform

Insert a single-panel free-surface or normal-discharge forcing waveform.

single-panel figure slot

Figure placeholder: convergence summary

Insert a single-panel error or runtime-versus-error figure once final reporting choices are fixed.

single-panel figure slot

Current achievements

- End-to-end source, terrain, Regional2D, coupling export and Local3D replay pathway implemented.
- G5/G6 Regional2D prefix equivalence checked for accepted forcing export.
- Production Local3D boundary, wall-function and timestep policies recorded for the G6 baseline.
- Real Kamaishi no-defence and simple-rigid-barrier replay acceptance captured.
- Reproducible convergence evidence and figure-provenance workflow established for C1A.

Next steps

1. Complete Regional2D convergence and select a defensible production forcing mesh.
2. Run Local3D spatial and temporal convergence once Regional forcing qualifies.
3. Calibrate uncertain source, terrain and boundary parameters using approved evidence.
4. Validate against arrival time, offshore waveform, run-up and inundation observations.
5. Extend to defence-class studies, then future FSI, damage modelling and optimisation.

Scan / contact

Scan for report, repository or project page.
Short link: `add-project-link`
Contact: `add-institutional-email`

QR and contact details are placeholders for the final poster export.

Evidence roadmap and final figure queue

Verification gate stack

- **G6:** hybrid theory, replay contract and real Kamaishi Local3D acceptance closed.
- **C1A:** Regional discretisation evidence active; current forcing waveforms not yet qualified.
- **Local3D convergence:** starts only after Regional forcing is defensible.
- **Validation:** arrival time, waveform, run-up and inundation comparison remain future work.

Final single-panel figure queue

- Tohoku source-to-Kamaishi corridor map.
- Hybrid 2D–3D workflow graphic.
- Regional forcing waveform.
- Convergence or runtime-versus-error summary.
- Local3D OpenFOAM replay visual.

Interpretation boundary

- Current figures are placeholders unless explicitly replaced by provenance-backed outputs.
- The poster may report implemented framework capability and accepted G6 evidence.
- It must not claim completed calibration, observational validation, FSI or defence performance conclusions.

Status note. This poster does not claim completed observational validation, calibration, full convergence closure, FSI or final defence-performance conclusions. It is a publication-quality baseline poster source for the implemented hybrid framework and current verification status.

Selected figure references. Encyclopaedia Britannica; Japan earthquake and tsunami of 2011; Encyclopaedia Britannica; n.d. NOAA National Centers for Environmental Information; ETOPO 2022 15 Arc-Second Global Relief Model; NOAA/NCEI; 2022. Saito et al.; Dispersion and Nonlinear Effects in the 2011 Tohoku-Oki Earthquake Tsunami; Journal of Geophysical Research: Oceans; 2014. Fujima, Masamura and Goto; Development of the 2D/3D Hybrid Model for Tsunami Numerical Simulation; Coastal Engineering Journal; 2002. Takase et al.; 2D–3D Hybrid Stabilized Finite Element Method for Tsunami Runup Simulations; Computational Mechanics; 2016.